

## 8 Proofs Involving Sets

Use the methods introduced in this chapter to prove the following statements.

**2. Prove that  $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ .**

*Proof.* First we show  $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ . Suppose  $a \in \{6n : n \in \mathbb{Z}\}$ . This means  $a = 6n$  for some integer  $n$ . Therefore  $a = 2(3n)$  and  $a = 3(2n)$ . From  $a = 2(3n)$ , it follows that  $a$  is a multiple of 2, so  $a \in \{2n : n \in \mathbb{Z}\}$ . From  $a = 3(2n)$ , it follows that  $a$  is a multiple of 3, so  $a \in \{3n : n \in \mathbb{Z}\}$ . Thus by definition of intersection of two sets, we have  $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ . Therefore  $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ .

Next we show  $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$ .

Suppose  $a \in \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ . By definition of intersection of two sets, this means  $a \in \{2n : n \in \mathbb{Z}\}$  and  $a \in \{3n : n \in \mathbb{Z}\}$ . Since  $a \in \{2n : n \in \mathbb{Z}\}$ , we know  $a = 2k$  for some  $k \in \mathbb{Z}$ . Thus  $a$  is even. Since  $a \in \{3n : n \in \mathbb{Z}\}$ , we know  $a = 3l$  for some  $l \in \mathbb{Z}$ . Since 3 is odd, if  $l$  were odd then  $3l$  would be odd; as  $a = 3l$  is even, it follows that  $l$  is even. Then  $l = 2j$  for some  $j \in \mathbb{Z}$ . So we have  $a = 3l = 3(2j) = 6j$ . Since  $j \in \mathbb{Z}$ , it follows that  $a \in \{6n : n \in \mathbb{Z}\}$ . Thus  $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$ .

Therefore, since  $\{6n : n \in \mathbb{Z}\} \subseteq \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$  and  $\{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\} \subseteq \{6n : n \in \mathbb{Z}\}$ , it follows that  $\{6n : n \in \mathbb{Z}\} = \{2n : n \in \mathbb{Z}\} \cap \{3n : n \in \mathbb{Z}\}$ .  $\square$

**4. If  $m, n \in \mathbb{Z}$ , then  $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$ .**

*Proof.* Suppose  $a \in \{x \in \mathbb{Z} : mn \mid x\}$ . Then  $a = mnk$  for some  $k \in \mathbb{Z}$ . So we have  $a = m(nk)$  and  $a = n(mk)$ . By definition of divisibility,  $m \mid a$  and  $n \mid a$ . Thus  $a \in \{x \in \mathbb{Z} : m \mid x\}$  and  $a \in \{x \in \mathbb{Z} : n \mid x\}$ . By definition of intersection of two sets, we have  $a \in \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$ . Therefore  $\{x \in \mathbb{Z} : mn \mid x\} \subseteq \{x \in \mathbb{Z} : m \mid x\} \cap \{x \in \mathbb{Z} : n \mid x\}$ .  $\square$

**6. Suppose  $A, B$  and  $C$  are sets. Prove that if  $A \subseteq B$ , then  $A - C \subseteq B - C$ .**

*Proof.* We prove the contrapositive: if  $A - C \not\subseteq B - C$ , then  $A \not\subseteq B$ . Assume  $A - C \not\subseteq B - C$ . Then there exists an element  $x \in A - C$  such that  $x \notin B - C$ . Since  $x \in A - C$ , we have  $x \in A$  and  $x \notin C$ . Since  $x \notin B - C$ , it follows that  $x \notin B$  or  $x \in C$ . Because  $x \notin C$ , it follows  $x \notin B$ . Since  $x \in A$  and  $x \notin B$ , by definition of a subset, we have  $A \not\subseteq B$ .  $\square$

**8. If  $A, B$  and  $C$  are sets, then  $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ .**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
 A \cup (B \cap C) &= \{x : x \in A \vee x \in B \cap C\} && \text{(definition of } \cup) \\
 &= \{x : x \in A \vee (x \in B \wedge x \in C)\} && \text{(definition of } \cap) \\
 &= \{x : (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} && \text{(distributive law)} \\
 &= \{x : x \in A \cup B \wedge x \in A \cup C\} && \text{(definition of } \cup) \\
 &= \{x : x \in (A \cup B) \cap (A \cup C)\} && \text{(definition of } \cap) \\
 &= (A \cup B) \cap (A \cup C) && \text{(definition of set-builder)}
 \end{aligned}$$

$\square$

**10. If  $A$  and  $B$  are sets in a universal set  $U$ , then  $\overline{A \cap B} = \overline{A} \cup \overline{B}$ .**

*Proof.* Observe the following sequence of equalities.

$$\begin{aligned}
 \overline{A \cap B} &= \{x \in U : x \notin A \cap B\} && \text{(definition of complement)} \\
 &= \{x \in U : \neg(x \in A \wedge x \in B)\} && \text{(definition of } \cap) \\
 &= \{x \in U : x \notin A \vee x \notin B\} && \text{(De Morgan's law)} \\
 &= \{x \in U : x \in \overline{A} \vee x \in \overline{B}\} && \text{(definition of complement)} \\
 &= \{x \in U : x \in \overline{A} \cup \overline{B}\} && \text{(definition of } \cup) \\
 &= \overline{A} \cup \overline{B} && \text{(definition of set-builder)}
 \end{aligned}$$

$\square$